



Complete System Overview

Every part of the winter services system, and how the parts fit together.

This system takes a customer from "what would it cost to winterize my boat?" all the way to a signed, paid, scheduled job — and gives Quest staff the tools to run the season from a phone. It is three pieces that share one set of pricing rules and one spreadsheet.

The three pieces

Piece	Who uses it	What it is
Quote page	Customers	A public web page where someone builds a winter quote for a boat, jet ski, golf cart or e-bike, and can email, print, sign and pay.
Staff console	Quest staff	A PIN-protected page for looking up quotes, taking payments, adjusting prices, sending emails, uploading photos and running the yard.
Backend + spreadsheet	Nobody directly	The Google Apps Script that prices, stores, emails and generates PDFs, plus the Google Sheet that holds every quote.

Why they agree with each other

Both the quote page and the backend price a quote using **the same rule set**, kept in one file. There is no second copy of the prices that could drift out of step. When a customer saves a quote, the server prices it again from their selections and stores its own figure — so a stale browser tab, or an old page left open across a price change, cannot lock in the wrong number. If the two ever disagree, the save still completes with the server's figure and an alert is emailed to service@ with **PRICE DRIFT** in the subject.

The customer's journey

Full step-by-step in **Document 3**. In outline:

- **Contact details first.** Name, phone and email are required before any pricing is shown. That button press is also where the customer agrees to the Terms — the acknowledgment sits next to it, with real links, and the version they agreed to is recorded with a timestamp.
- **A row is logged immediately.** The moment they pass that gate, a row appears on the **Quote Started** tab. If they wander off, you still know who reached the pricing.
- **They build the quote.** The questions asked depend on the unit type — a boat is asked about engines, systems, storage and shrinkwrap; a jet ski is asked far less; golf carts and e-bikes are close to flat-rate.
- **They review and act.** Print, email themselves a copy, sign, or pay a deposit.
- **They can come back.** A quote reloads with quote number and last name, with every selection restored. If they start again with the same email and last name, the page offers to continue the quote they already had rather than making a second one.

What staff do

Full detail and permission levels in **Document 2**. The console covers: looking up a quote by number or partial last name; recording payments, deposits and refunds; editing or removing line items; pricing a service the

customer asked about; applying late fees; correcting dimensions, motor counts and storage location; uploading condition photos and signed contracts; previewing and sending customer emails, individually or to everyone; printing yard sheets and the haul-out list; managing staff accounts; and restoring from a backup.

Staff changes are never lost when a customer edits their quote

Every staff change is recorded as an *operation* — "discount this line", "price this request", "we measured the beam at 9.4 feet" — not just baked into the numbers. When a customer reloads and re-saves, the server rebuilds their quote cleanly from their selections and then replays every staff change on top. Without this, a customer opening their quote would quietly erase your discounts.

Correcting what the customer told us

Customers get things wrong — the wrong beam, the wrong number of motors, the wrong storage. The console's **Unit details & storage** card handles all three. You change a value, press **Preview the change**, and the server re-prices and shows you a line-by-line before/after with the new total. Nothing is written until you press **Apply**.

- Corrections are stored separately from what the customer selected, so their next save cannot undo them — and you can always still see what they originally told you.
- A beam that outgrows its storage class is **flagged, never moved**. Relocating a customer's boat is a conversation, not something the software should decide.
- Moving storage physically moves the quote to the right tab and re-prices it, because inside and outside cost different money.
- Motors are corrected as a set — type, count and service level together — because a boat can have several of one type but never a mix.

Money

Payments are an **append-only ledger**. Nothing is ever deleted; a refund is recorded as a negative payment. Balance is always total minus paid, and it is never clamped at zero — if a customer has overpaid, the system shows a **CREDIT** in the sheet, on the PDF, in emails and on their own page, rather than pretending the balance is zero.

Once any payment exists the quote **locks**. The customer's page becomes read-only with a gold banner, the wording changes from Quote to Invoice everywhere, and the server refuses to let a stale browser tab overwrite a paid quote. Staff can still adjust a locked quote from the console — that is how corrections after a deposit get made.

Emails

Every customer email is drafted by one shared builder, so the preview you see is exactly what sends. **Only two emails ever send by themselves:**

Automatic email	When	Who gets it
Quote reminder	10 days after a quote is saved or emailed with no signature and no payment. Runs daily at 9:00am Central.	Customers with a real quote. Never anyone who only reached the pricing and stopped.

Automatic email	When	Who gets it
Finish my quote	Once, roughly 24–47 hours after someone passes the contact gate and walks away. Runs daily at 12:15pm Central, chosen to land over lunch.	Only people still sitting on the Quote Started tab. Saving, printing, emailing or paying moves them off it.

Everything else — receipts, "we have your unit", the seasonal notes, "you're up next", late-fee warnings, dimension-change notices — requires a staff member to click, preview and confirm. The 1st and 15th balance reports go to Chris only, never to customers. Both automatic emails are written to the Activity Log as well, so "did we contact this customer?" has one answer no matter who sent it.

The two seasonal announcements

These are the only emails that have a send-to-all version, and they bracket the season. The **end-of-season note** goes out in the autumn: the season is winding down, here is a last chance at detailing or winter work, and when would you like to come out? It will not offer a detail quote to somebody who has already asked for one. The **spring relaunch alert** is its counterpart in the other direction — anything you want done before it goes back, and when would you like it? Both put the customer's answer into the same haul-out timing record, so the late retrieval surcharge applies from either.

Alongside them, **"you're up next"** comes in two directions. In autumn it means we are about to take the unit out; in spring, about to put it back. They are separate emails rather than one with the words flipped, because the useful thing to say differs: in autumn the last easy moment to add work is now, while we have it.

The season-done question

Customer emails carry a three-option question — done now, done on a set date, or "I'll call" — but **only once a deposit or payment exists**. Asking someone to book a haul-out before they have put money down is asking them to schedule work they have not agreed to buy. A date after Nov 15 automatically applies the late retrieval surcharge, priced from that quote's own stored figure so it stays right year to year. Staff can record the same answer from the console for anyone who phones in.

Sending to everyone

A send-to-all is the one action in the system that cannot be undone or narrowed after the fact, so it is built to be checked first. Choosing the email does not send it: the console reads the sheet and reports who would receive it, broken down by storage area, plus how many quotes have no email address and whether the number is large enough to hit Google's daily sending limit. It also renders the real email for the first person on the list. Sending is a second, separate, confirmed click.

People who started a quote and walked away are never on that list. They are leads, not customers, and a "your boat" email to a stranger — possibly a competitor — is the failure the exclusion exists to prevent. It is enforced in one shared piece of code used by both the console and the spreadsheet menu, and there is an automated test that builds a spreadsheet with a lead on it and checks the lead is not in the result.

The list is also a picker. Everyone arrives ticked — it is a send-to-all, and the point is to drop the occasional customer, not to build a list from scratch — and you untick whoever should not get it, individually or a whole storage area at a time. Whatever you send back can only make that list *shorter*: the recipients are decided on the server before your selection is applied, so a ticked box cannot add somebody the rules already excluded.

Keys, slips, and chasing what we do not know

A haul-out needs two things the customer often has not given us: where the keys will be, and which slip the boat is in. Both are editable on the console, and both are stored separately from what the customer selected, so their next save cannot undo a correction.

If either is missing, the haul-out "you're up next" email asks for it directly. It only asks for what applies — an e-bike has no keys — but a trailer is **not** taken to mean the boat is out of the water, because plenty of customers store the trailer with us and keep the boat in a slip all season. When we have both, the email says them back so a stale key location gets corrected before a truck goes out.

Photos and contracts

Condition photos are uploaded from the console — camera or gallery — and filed into Winter and Spring folders per quote, so there is a record of how a unit arrived. Signed contracts upload the same way. Both are stored in Drive as "anyone with the link can view", which is what lets staff open them without a Google login. See **Document 4**.

What runs on a timer

Time (Central)	What happens
9:00am daily	Ten-day quote reminders to customers who have not signed or paid.
12:15pm daily	"Finish my quote" to people who stopped at the contact gate ~24h ago.
6:00pm daily	Full spreadsheet backup emailed to Chris as an .xlsx file.
7:00am	Unpaid balance report — internal only, to Chris.

Google fires these within roughly a quarter-hour of the stated time; that cannot be tightened and does not matter for any of them.

Not built yet

Item	Status
Adobe Sign web form	Waiting on the contract being finalised. Until the URL is set, customers see a "signing almost here" placeholder instead of a broken button. Everything else in the couch-to-paid path already works.
BiT DMS work orders	On hold. No public API was found; the next step is asking BiT whether they offer one or accept a file import.
Import of last year's selections	Needs a sample workbook to map the columns. The design already supports it, because quotes store the customer's selections rather than just prices.
Text-message copies of emails	Blocked on carrier registration (roughly \$20–65 once, \$50–60 a year, about a month to approve).

One thing to decide before the season opens

The season dates and prices still read **2025–2026** (pay by Nov 15 2025, storage Oct 15 2025 – Apr 15 2026) while the spreadsheet is named 2026-2027. Rolling the season over is a single edit in one place that updates the customer page and the backend together, but it has not been done yet and it changes what customers are told. Worth doing deliberately rather than discovering it mid-quote.